

Launch from the ISS

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1 History

To offer more frequent opportunities and moderate environments for small satellite launches into space, a new mission concept capable of deploying small satellites using the Japanese Experiment Module (JEM) Small Satellite Orbital Deployer (J-SSOD) of the International Space Station (ISS) in low-Earth orbit (~400 km) was developed by the Japan Aerospace and Exploration Agency (JAXA).

This capability was first demonstrated by the successful deployment of five CubeSats (four 1U and one 2U satellites) from the J-SSOD on October 4, 2012. These CubeSats, containing pressurized cargo softly stowed in a bag inside an H-IIB Transfer Vehicle (HTV)-3, were launched by H-IIB rocket from the JAXA Tanegashima Space Center. After being installed onto the JEM Remote Manipulator System (JEMRMS) by the ISS crew and positioned by both crew and ground controller commands to the JEMRMS, they were successfully deployed to their planned orbit.

After this success, by 2018, more than 200 microsattellites had been deployed from the JEM using several deployment mechanisms such as the J-SSOD, NanoRacks CubeSat Deployer (NRCSD) developed by NanoRacks LLC, and CYCLOPS [or Space Station Integrated Kinetic Launcher for Orbital Payload Systems (SSIKLOPS)] developed by NASA. Launching microsattellites from the ISS became one of the major launch opportunities for CubeSat developers and users. As a result of the use of the International Space Station, potential developers of small satellites have increased their use of space station deployers, and universities, companies, and other nontraditional space users are realizing affordable access to space.

2 Mission overview [1]

As a new way of transferring CubeSats and other small satellites into space, small satellite deployment missions from the JEM by using a robotically manipulated launcher system were developed. Satellite developers or users can exploit this unique capability with the JEM Airlock and JEMRMS (Fig. 1).

The JEM Airlock is a unique feature that moves equipment from inside the ISS to outside space but is not used for the crew's extravehicular activity. There is a slide table in the JEM Airlock, which can be extracted both inside and outside the JPM cabin. The J-SSOD is installed on the Multipurpose Experiment Platform (MPEP)

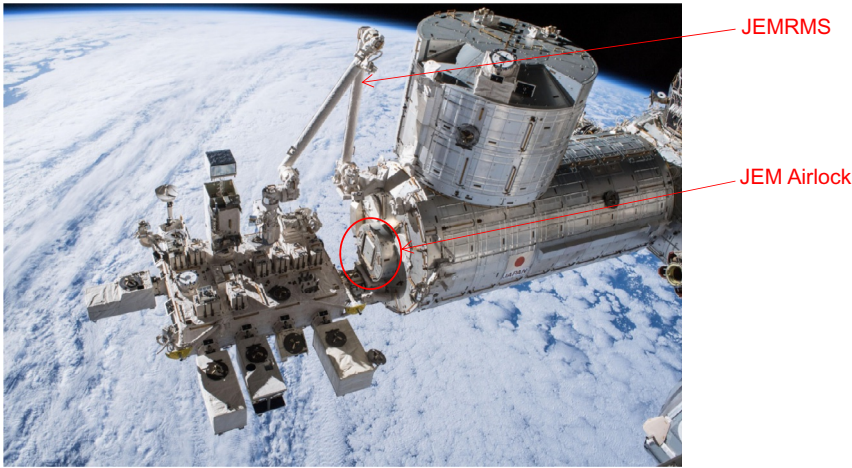


Fig. 1 JEM outside view.

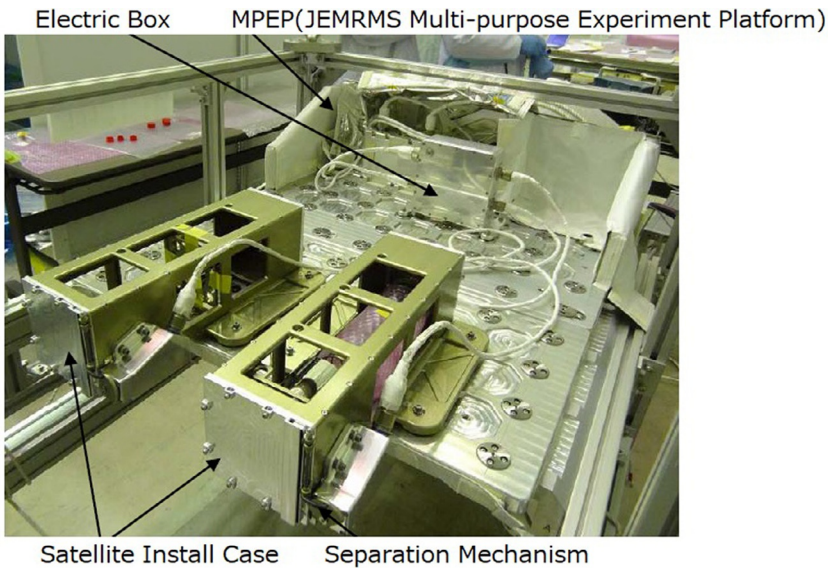


Fig. 2 J-SSOD and MPEP.

(Fig. 2). The MPEP is a platform that acts as an interface between operations inside and outside the ISS, and the J-SSOD mechanism is installed on this platform.

J-SSOD missions are performed using the following steps as a typical scenario to deploy satellites from the JEM. A typical CubeSat deployment scenario movie can be viewed at: http://issstream.tksc.jaxa.jp/iss2/video/jssod_170530.mp4.

1. Satellites are installed into the Satellite Install Case as launched from Earth configuration.
2. The Satellite Install Case is packed in the ISS Cargo Transfer Bag (CTB) with a cushion, launched as a pressurized cargo, and delivered to the ISS by a resupply vehicle like an HTV or other commercial Cygnus or Dragon.
3. In the ISS, once the Satellite Install Cases with prepacked small satellites are received onboard, crew members unpack and install them onto the MPEP, which is already installed onto the JEM Airlock Slide Table, along with other equipment related to the J-SSOD mechanism as shown in [Fig. 3](#).
4. The J-SSOD on the MPEP is passed through the JEM Airlock to outside the ISS and is grappled and handled by the JEMRMS, as shown in [Fig. 4](#).
5. A JAXA robotics specialist ground flight controller team called “KIBOTT” maneuvers the JEMRMS to the appropriate deployment orientation.

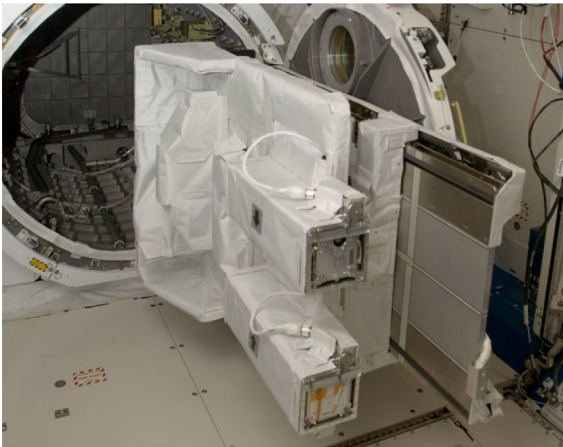


Fig. 3 J-SSOD installation in the JEM Airlock [\[2\]](#).

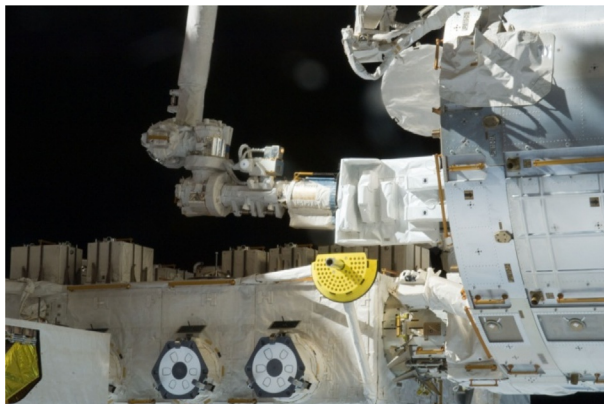


Fig. 4 The J-SSOD is transferred to the deploy position by the JEMRMS [\[2\]](#).



Fig. 5 Satellites are deployed into space [2].

6. The satellites are deployed from the Satellite Install Case of the J-SSOD into space using commands from the crew or ground flight controller, as shown in [Fig. 5](#). The Space Station Remote Manipulator System and other ISS cameras may be used to support video monitoring of satellite deployment as necessary. Crew members also monitor the deployed CubeSats and take images as shown in [Fig. 6](#).
7. The J-SSOD is returned to the airlock and stowed inside the ISS.

Deployment of CubeSats or other small satellites requires a specific JEMRMS orientation (i.e., safe deployment cone) to avoid ISS contact. [Table 1](#) shows typical parameters of the deployment orbit [3, 4].

The J-SSOD is capable of deploying not only 1U to 3U CubeSats, but also a 50 kg class microsatellite of size $55\text{ cm} \times 55\text{ cm} \times 35\text{ cm}$. The J-SSOD is capable of

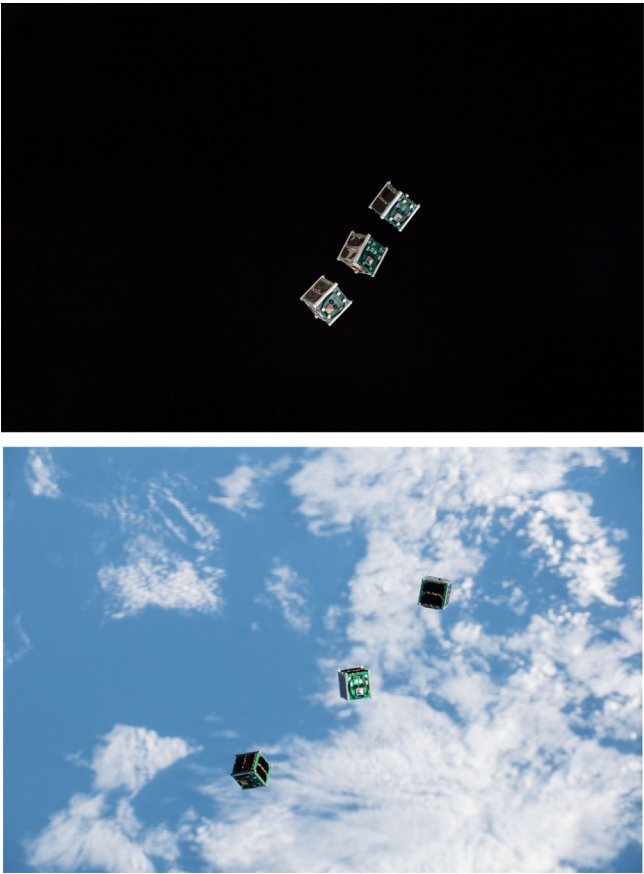


Fig. 6 Satellites are monitored after deployment.

Table 1 Deployment orbit parameters (typical).

Parameters	Specification
ISS orbit	1. Altitude at approx. 380–420 km (typically around 400 km) 2. Inclination at 51.6 degrees
Deployment direction	Nadir-aft, 45 degrees from the nadir with respect to the ISS Body Coordinate System
Deployment velocity	1.1–1.7 m/s for 3U satellite
Orbiting lifetime	If the ballistic number (BN) is 100, about 100–250 days depending on the deployment altitude and atmospheric conditions

deploying 12U ($3U \times 4$) in its single mission. Following the J-SSOD, NRCSD, and CYCLOPS were also developed. Details of these other two launch systems will be described in [Section 4](#).

3 Advantage of launching from the ISS [1]

As a result of the use of the International Space Station, potential developers of small satellites have increased their use of space station deployers, and universities, companies, and other nontraditional space users are finding affordable access to space. The major advantages of launching from the ISS using deployers like J-SSOD, NRCSD, and CYCLOPS are described in the following sections.

3.1 More frequent launch opportunities

Previously, CubeSats or small satellites of a certain class were launched by rockets as piggyback satellites only. If a rocket could launch extra weight in addition to its main satellite, piggybacked smaller satellites were given the opportunity to be placed into orbit after the main satellite was successfully deployed. In a usual piggyback launch, satellite development should be completed on time once its launch schedule is fixed. On the other hand, launch opportunities from the ISS are provided by resupply vehicles like HTV, Cygnus, or Dragon. This means that there is an opportunity to allocate vehicles less than 10 times per year; therefore increased flexibility is expected in selecting the launch vehicle by considering the satellite development schedule. Also, other options are available, e.g., choosing the best timing for the small satellite's ejection without affecting the main satellite's timing.

3.2 Moderate launch environment

All satellites must pass space environment tests to confirm that they will survive the hard launch environment in space. A vibration test that simulates vibrations experienced during launch subjects the satellite to quite a strong level of vibration. Piggyback satellites are required to pass this severe test. For launch from the ISS, since the satellites are packed in a CTB with cushions and launched in a "pressurized" cargo, the vibration environment is more moderate than a usual piggyback launch. This relaxed condition is more applicable for college students or a less costly satellite project.

3.3 Checkout after launch from Earth and before deployment from the ISS

One of the issues with piggyback satellites is that once they are launched, it is uncertain whether the satellite is still in good working order after enduring launch vibration. An additional advantage of launching from the ISS is that the satellites can be checked out by the crew in the pressurized ISS cabin to confirm their integrity with additional assessment/coordination to assure compatibility with ISS operation and safety.

4 Launch history and lessons learned

As of October 2018, the J-SSOD deployed 34 CubeSats and one 50 kg class microsatellite, that were developed in several countries as shown in Table 2 and Fig. 7.

NRCSD development started in February 2014 and NanoRacks LLC has experienced NRCSD operations many times (Fig. 8). Recently, NanoRacks LLC has also

Table 2 History of J-SSOD satellite deployment.

Mission #	Deployment date	Number of satellites	Satellite size
1	October 2012	5	1U × 4, 2U × 1
2	November 2013	4	1U × 3, 3U × 1
3	February 2015	1	1U × 1
4	September 2015	2	3U × 2
5	April 2016	1	50 kg class × 1
6	December 2016 and January 2017	7	1U × 3, 2U × 2, 3U × 2
7	July 2017	5	1U × 5
8	May 2018	3	1U × 2, 3U × 1
9	August 2018	3	1U × 3
10	October 2018	3	1U × 1, 2U × 2

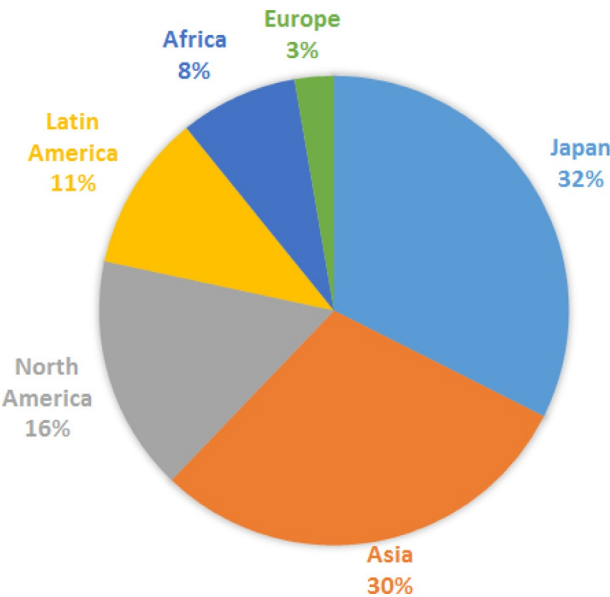


Fig. 7 History of J-SSOD satellite deployment.

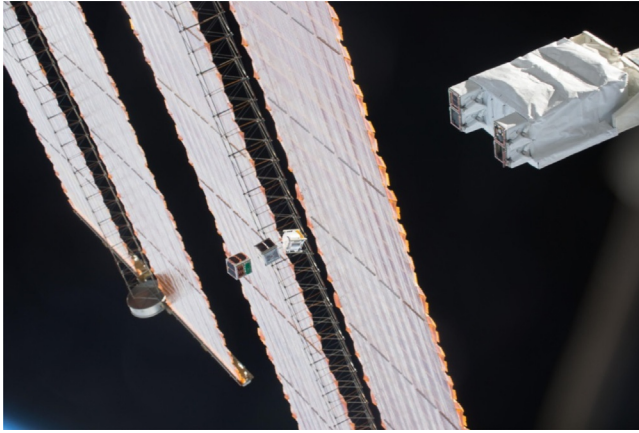


Fig. 8 Deployment of NRCSat.



Fig. 9 Deployment of RemoveDebris.

developed a microsat deployer called “KABER” and deployed a European satellite called “RemoveDebris” (Fig. 9) with its size approximately at $65\text{ cm} \times 65\text{ cm} \times 72\text{ cm}$. This is the maximum size ever deployed from the ISS through the JEM Airlock, which fully utilized the maximum capable volume of the JEM Airlock. Spinsat was successfully deployed by NASA/CYCLOPS in 2014 using both JEMRMS and Small Fine Arm (Fig. 10).

The lessons learned from these types of launches have identified key requirements that need to be met. It can be seen in Fig. 11 that the design and verification of deployable mechanisms such as antennas and rechargeable batteries are major issues from an ISS safety point of view. Especially for Li-ion rechargeable batteries, screening tests on flight cells to show their integrity are required, such as random vibration and vacuum leak tests. Also the battery circuit requires protection devices for short circuit, overcharge, and overdischarge.

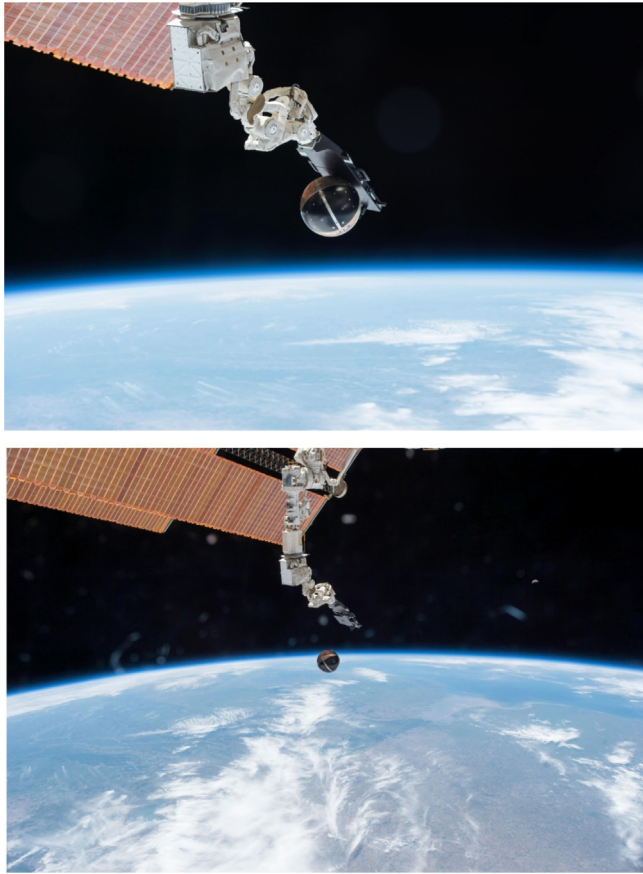


Fig. 10 Deployment of SpinSat.

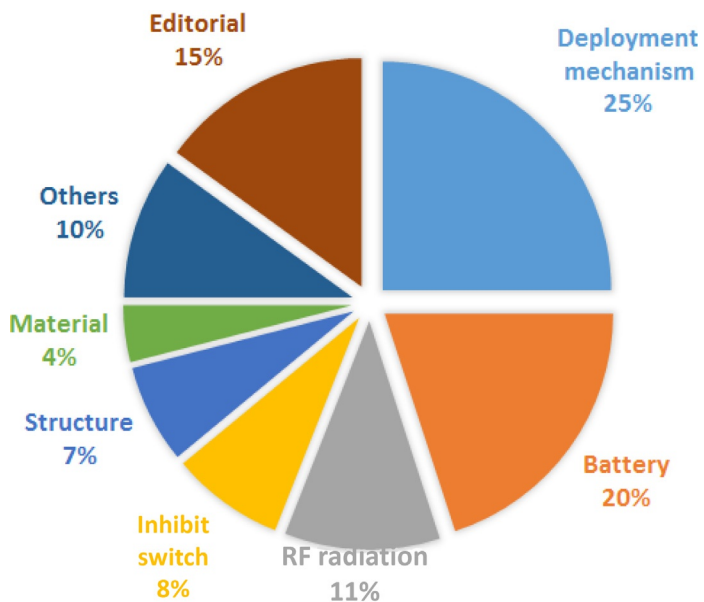


Fig. 11 Category of typical comments at a safety review for a J-SSOD satellite.

References

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